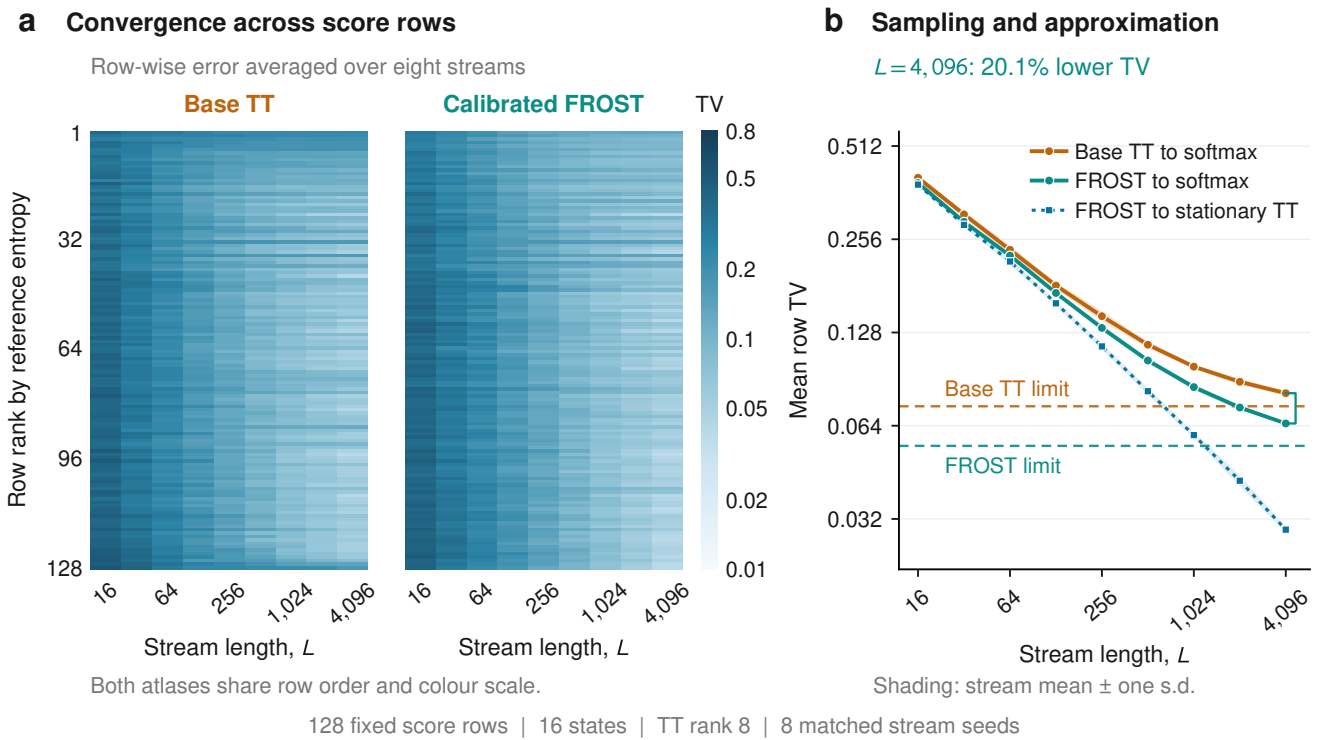

a
Convergence across score rows
Row-wise error averaged over eight streams
Base TT
Calibrated FROST
TV
0.8
0.5
0.2
0.1
0.05
0.02
0.01
Row rank by reference entropy
1
32
64
96
128
16
64
256
1,024
4,096
Stream length, L
b
Sampling and approximation
L = 4,096: 20.1% lower TV
Base TT to softmax
FROST to softmax
FROST to stationary TT
Mean row TV
0.512
0.256
0.128
0.064
0.032
Base TT limit
FROST limit
Shading: stream mean ± one s.d.
Both atlases share row order and colour scale.
128 fixed score rows | 16 states | TT rank 8 | 8 matched stream seeds